

## TAKEIN STUDIO · TECHNICAL HANDBOOK VAULT

# The AI Web Developer's Playbook

Volume 1: AI-Native Foundations, Claude Code Mastery, 21st.dev Component Libraries, Framer Motion & Prompt Engineering Systems

**FIELD GUIDE SYLLABUS & MODULES INCLUDED:**

- 1. Claude Code Autonomous Terminal Developer
- 2. 21st.dev LEGO Blocks & Motion Psychology
- 3. The Golden Website Prompt Framework
- 4. 8-Stage Conversion Architecture
- 5. \$100k SaaS Flagship Master Prompt
- 6. Gemini 3 C.R.E.F. Formula & Cheatsheet
- 7. The 6 Levels of AI Independence
- 8. Antigravity IDE & Context Engineering (@)
- 9. Model Routing Matrix & 12 Architectures
- 10. Dual-Track Deployment (GUI vs CLI)
- 11. Conversion Design & Color Psychology
- 12. 3-Page Flagship SaaS Capstone Project

## 01. What is Claude Code & The AI Web Development Paradigm Shift

Most web developers mistakenly believe Claude Code is "just another AI chatbot". **It is not.** Claude Code is an autonomous AI software engineer that operates directly inside your local computer's terminal. It inspects file trees, executes shell commands, runs test suites, and edits multi-file React architectures without human copy-pasting.

### THE AI-NATIVE DEVELOPER DEFINITION:

Claude Code combines a **Senior Product Manager + Lead UI Architect + Full-Stack Engineer + QA Auditor** running directly inside your CLI environment. You act as the Director; the AI agent acts as the execution team.

#### TRADITIONAL WORKFLOW (2 – 8 WEEKS)

- Figma wireframing & prototype client approval (5–10 days)
- Manual Tailwind CSS component styling & layout (7–14 days)
- Backend API integration & database setup (7–14 days)
- Cross-browser testing, debugging & Vercel deployment (3–7 days)

Total Time: 14 – 60 Days | Cost: \$5,000–\$25,000

#### TAKEIN AI-NATIVE WORKFLOW (HOURS)

- Golden Website Prompt + 21st.dev Component Library (15 mins)
- Claude Code terminal execution & Framer Motion (45 mins)
- Builder → Reviewer subagent verification & polish (30 mins)
- GitHub Desktop GUI or Vercel CLI 1-click launch (10 mins)

Total Time: 2 – 4 Hours | Cost Reduction: 90%

#### AIWEBDEV PROMPT 1.01

#### TERMINAL CLI (BUILD WITH AI · SETUP COMMAND)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

```
# Step 1: Install Anthropic's Claude Code CLI globally
npm install -g @anthropic-ai/claude-code
```

```
# Step 2: Initialize workspace & launch autonomous agent
mkdir my-ai-website && cd my-ai-website
claude
```

### YOUR TASK · 01.1 — ENVIRONMENT INITIALIZATION

### EXERCISE

- 1. Verify Node.js v18+ is installed on your operating system.
- 2. Execute `claude` in your terminal and complete the Anthropic OAuth browser authentication.
- 3. Type `/help` inside the Claude Code prompt interface to view built-in commands.

DELIVERABLE: Type `claude --version` in terminal. Record output version: \_\_\_\_\_

## 02. Essential Toolkit: 21st.dev UI Components & Framer Motion

Why do elite SaaS websites (Stripe, Linear, Vercel) feel insanely premium while amateur sites feel cheap? It is **not** the font choice or background color. It is **Motion Psychology** and **Modern Component Primitives**.

### Framer Motion (Visual Authority)

Static web pages feel dead. Motion creates micro-dopamine triggers as users scroll, giving the impression of an expensive native desktop application.

- **Smooth Scroll Reveals:** Section headers slide up 20px on entry.
- **Hover Scale & Glow:** Feature cards scale 1.02x with radial light.
- **Staggered Entrance:** Grid elements appear sequentially (100ms offset).

### 21st.dev (Modern UI LEGO Blocks)

21st.dev is the open-source library of modern React + Tailwind primitives engineered by top designers specifically for AI code generation.

- Pre-styled Glassmorphic Navigation Bars
- Interactive Pricing Tables with Monthly/Yearly Toggles
- Animated Testimonial Marquees & Grid Layouts

#### AIWEBDEV PROMPT 1.02

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

#### NEXT.JS / TAILWIND (BUILD WITH AI · PACKAGE INSTALLATION PROMPT)

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

Install the required animation and utility dependencies for modern 21st.dev components:

```
npm install framer-motion lucide-react clsx tailwind-merge cva
```

Create a reusable utility file at `lib/utils.ts` with `cn()` helper function for dynamic Tailwind class merging.

#### // Framer Motion Staggered Card Component Blueprint

```
import { motion } from 'framer-motion';
```

```
export const FeatureCard = ({ title, description, index }: { title: string; description: string; index: number }) => (
```

```
  <motion.div
```

```
    initial={{ opacity: 0, y: 24 }}
```

```
    whileInView={{ opacity: 1, y: 0 }}
```

```
    transition={{ duration: 0.5, delay: index * 0.12 }}
```

```
    whileHover={{ scale: 1.02, transition: { duration: 0.2 } }}
```

```
    className="p-6 bg-slate-900 border border-slate-800 rounded-xl shadow-xl hover:border-orange-500/50"
```

```
  >
```

```
    <h3 className="text-lg font-bold text-white mb-2">{title}</h3>
```

```
    <p className="text-slate-400 text-sm">{description}</p>
```

```
  </motion.div>
```

```
);
```

#### 💡 PRO TIP · MOTION THRESHOLD RULE:

Never use animation durations over **0.6s** for UI elements. Slow animations frustrate visitors and destroy conversion rates. Keep entrance transitions between **0.3s and 0.5s** with `easeOut` timing curves.

## 03. The Golden Website Prompt Framework

Vague prompts produce generic, ugly HTML output that looks like a 2012 Bootstrap site. To force Claude Code or Antigravity to generate Apple/Stripe-grade frontend code, you must use the **Golden Website Prompt Architecture**.

### ❌ Vague Prompt (Low Quality Output)

Build me a modern AI agency landing page with Tailwind CSS.

Result: Plain white background, default blue buttons, no responsive viewports, no animations, boring text blocks.

### ✅ Golden Prompt Architecture (5 Core Pillars)

1. **Persona & Target Audience:** Who is visiting?
2. **Aesthetic Inspiration:** Linear + Apple + Stripe.
3. **Tech Stack:** Next.js App Router, Tailwind, Framer Motion.
4. **Component Standards:** 21st.dev glowing cards, high contrast.
5. **Conversion Goal:** High-intent lead capture.

#### AIWEBDEV PROMPT 1.03

#### COPY & PASTE INTO CLAUDE CODE (THE GOLDEN WEBSITE MASTER PROMPT)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

Act as a Lead UI/UX Designer & Senior Next.js Developer. Build a high-converting website.

PROJECT GOAL: Create a flagship website for [INSERT CLIENT / PRODUCT NAME].

TARGET AUDIENCE: Founders, CTOs, and tech startup leaders looking for premium solutions.

DESIGN AESTHETIC: Dark theme (#0F172A primary background), Linear + Apple + Vercel aesthetic, TakeIN Orange (#FF6B00) primary call-to-action buttons, high typography contrast.

##### TECHNICAL REQUIREMENTS:

- Next.js 14 App Router, TypeScript, Tailwind CSS, Framer Motion.
- Fully responsive across mobile (375px), tablet (768px), and desktop (1280px+).
- Use Framer Motion for smooth section entrance reveals and card hover micro-interactions.
- Implement 21st.dev modern component primitives (hero badge, glowing border cards, interactive FAQ).

STRUCTURE: Hero Section → Social Proof Logo Wall → Problem → Solution → Features Grid → Testimonials → Pricing → CTA → Footer.

### YOUR TASK · 03.1 — PROMPT REFACTORING PRACTICE

### WORKBOOK

Refactor this bad prompt: "Make a restaurant website with a menu." using the 5 Golden Pillars:

---

---

---

## 04. Universal 8-Stage Website Structure & Storylines

Every high-converting website follows a psychological narrative that takes a cold visitor from initial interest to taking immediate action. Never throw random components onto a page.

### THE 8-STAGE UNIVERSAL HIGH-CONVERTING PIPELINE:

Stage 1: Hero (Hook & Clear Value Prop) → Stage 2: Social Proof (Logos/Numbers)  
→ Stage 3: Problem (Agitate Visitor Pain) → Stage 4: Solution (Your Transformation)  
→ Stage 5: Features Grid (How It Works) → Stage 6: Testimonials (Real Case Studies)  
→ Stage 7: Final Offer CTA (Risk Reversal) → Stage 8: Conversion Footer

### THE 3 CORE CONVERSION STORYLINE FRAMEWORKS

#### 1. Problem-Solution

Best for SaaS tools & utilities solving immediate headaches.

1. Agitate current manual struggle
2. Expose why old tools fail
3. Introduce AI automation
4. Show measurable speed gains
5. Primary trial CTA

#### 2. Dream Outcome

Best for high-ticket agencies, consulting & coaching.

1. Paint the desirable future state
2. Identify the strategic bottleneck
3. Present your proprietary system
4. Case study proof metrics
5. Book strategy call CTA

#### 3. Founder Story

Best for personal brands, creators & indie hackers.

1. Personal origin frustration
2. The breakthrough discovery
3. Building in public
4. Community & user praise
5. Join movement CTA

### YOUR TASK · 04.1 — MAP YOUR CLIENT'S STORYLINE

### ANALYSIS

Select one storyline framework above for your current website project and outline Stage 1 (Hero Outcome) vs Stage 3 (Visitor Pain Point):

**STAGE 1 HERO HEADLINE:**

**STAGE 3 VISITOR PAIN POINT:**

## 05. Specialized Website Prompt Vault

Ready-to-use production prompts tailored for specific commercial business models. Copy directly into your AI development agent:

### AIWEBDEV PROMPT 1.04

### PRODUCTION PROMPT (1. SAAS PLATFORM PROMPT (STRIPE + LINEAR VIBE))

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Build a modern SaaS landing page for an AI document analysis platform.

Style: Dark slate background (#0F172A), glowing border feature cards, subtle grid line overlay.

Sections:

- Hero: Glowing pill badge ("AI Document Intelligence 2.0"), gradient headline, dual CTAs ("Start Free Trial" orange button, "View Live Demo" ghost button).
- Interactive Pricing: Monthly/Yearly toggle with 3 tiers (Starter \$29, Pro \$79, Enterprise \$199).
- Framer Motion stagger entrance on 3-column feature grid.

### AIWEBDEV PROMPT 1.05

### PRODUCTION PROMPT (2. MINIMAL LUXURY AGENCY PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Create a high-end luxury creative agency website.

Style: Minimalist high contrast (Black & Pure White), bold editorial typography (Playfair / Inter mix), smooth page transitions.

Sections:

- Hero: Full-screen video/image placeholder with overlay text "We Engineer Digital Dominance".
- Selected Work: 2-column image grid with hover zoom effect and project metadata overlay.
- Contact: Sleek lead capture form with custom styled input fields.

### AIWEBDEV PROMPT 1.06

### PRODUCTION PROMPT (3. PERSONAL BRAND & FOUNDER PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Create a personal brand website for a tech founder & investor.

Sections: Hero (Headshot + Cinematic Tagline), Achievements Counter (e.g. "\$10M+ Raised", "3x Exits"), Featured Articles, Testimonials Carousel, Contact Form.

Style: Warm dark mode, crisp typography, subtle gold/orange accent highlights.

## 05. Flagship \$100k-Level SaaS Master Prompt

Use this comprehensive prompt when building high-value client production sites. It specifies architectural standards, design tokens, error handling, and animation physics.

### AIWEBDEV PROMPT 1.07

### FULL SPECIFICATION (FLAGSHIP \$100K SAAS MASTER PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Act as an Elite Principal Software Architect & Lead UI Designer at TakeIN Studio.

Create a world-class \$100k-level SaaS web experience.

#### DESIGN ARCHITECTURE & AESTHETICS:

- Palette: Background #0F172A (Deep Slate), Accent #FF6B00 (TakeIN Orange), Neutral #F8FAFC, Borders #1E293B.
- Typography: Inter font family, modular 1.25x type scale, bold high-contrast display headings.
- Visual Depth: Subtle radial gradient spotlights behind hero elements, glowing border card hover states.
- Motion Physics: Framer Motion spring physics (stiffness 300, damping 24) on all interactive buttons and cards.

#### COMPONENT ARCHITECTURE:

1. Navbar: Glassmorphic sticky header with blur backdrop, brand logo mark, navigation links, and "Get Started" CTA.
2. Hero: High-converting headline, sub-headline, trust badge, primary CTA button, and interactive product screenshot dashboard mockup with glowing border.
3. Social Proof: Animated infinite logo marquee featuring top tech brand logos.
4. Problem vs Solution: Interactive 2-column comparison card displaying "Traditional Way" vs "AI-Powered Way".
5. Features: 3x3 Bento Grid layout with custom icons, title, description, and hover scale effects.
6. Testimonials: 3-column testimonial grid with customer avatar images, star ratings, names, and verified badges.
7. Pricing Table: Monthly / Yearly toggle switch with 3 pricing tiers (Free, Pro, Enterprise) highlighting "Most Popular" tier in orange.
8. FAQ Accordion: Smooth animated collapsible questions and answers.
9. CTA Section: Full-bleed banner with high-urgency headline and lead input form.
10. Footer: 4-column link layout with copyright, legal disclaimers, and social icons.

#### CODE QUALITY:

- 100% TypeScript type safety. Clean component modularization in `components/ui/`. Zero console warnings or missing imports. Responsive across mobile (375px), tablet (768px), and desktop (1280px).

## 06. Mastering Gemini 3: Deep Think & The C.R.E.F. Method

Gemini 3 brings **Deep Think multi-step reasoning** and 1M+ token context windows. To unlock maximum precision, construct every prompt using the **\*\*C.R.E.F. Formula\*\***:

| C.R.E.F. ELEMENT         | STRATEGIC PURPOSE                         | EXAMPLE IMPLEMENTATION                                                           |
|--------------------------|-------------------------------------------|----------------------------------------------------------------------------------|
| <b>C - Context</b>       | Provide business background & constraints | "Building a Next.js 14 lead capture site for a dental clinic in Delhi."          |
| <b>R - Role</b>          | Assign specific AI expert persona         | "Act as a Senior Conversion Rate Optimization Specialist & Lead Engineer."       |
| <b>E - Explicit Task</b> | State the exact step-by-step deliverable  | "Generate a 3-step booking form component with Zod input validation."            |
| <b>F - Format</b>        | Define output structure & syntax          | "Output ready-to-copy TSX code in a single file block without markdown chatter." |

AIWEBDEV PROMPT 1.08

GEMINI 3 PRO DEEP THINK (FULL C.R.E.F. PROMPT BLUEPRINT)

|                             |                       |
|-----------------------------|-----------------------|
| USE WHEN                    | RECOMMENDED ROLE      |
| Following Volume I Workflow | Standard Builder      |
| CONTEXT TO ATTACH           | DO NOT ATTACH         |
| @project                    | .env.local            |
| RUN INSIDE                  | NEXT                  |
| Antigravity Agent           | Verify Implementation |

[CONTEXT]: I am launching an AI web development agency targeting local healthcare practices.  
[ROLE]: Act as a Senior UX Architect and Full-Stack Next.js Developer.  
[EXPLICIT TASK]: Create an interactive appointment booking component with real-time slot selection, form validation, and instant WhatsApp notification trigger.  
[FORMAT]: Provide production-ready React TSX code using Tailwind CSS and Lucide React icons.

✓ **VERIFICATION PROTOCOL · DEEP THINK EVALUATION:**

When using Gemini 3 Deep Think mode, check the reasoning trace before accepting code. Ensure it has evaluated mobile touch targets (48px minimum) and handled form loading states.



## 06. Gemini 3 Developer Cheatsheet: Do's and Don'ts

Avoid common prompting traps when directing Gemini 3 for code generation, copywriting, and repository analysis:

| TASK AREA           | ❌ COMMON MISTAKE (DON'T DO THIS)                                                               | ✅ AI-NATIVE PROMPT (DO THIS)                                                                                                      |
|---------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Code Generation     | "Write a contact form." (Generates basic unstyled HTML input tags).                            | "Build a modern contact card component using Tailwind CSS, Framer Motion reveal, and Zod email schema validation."                |
| Copywriting         | "Write a headline for my software." (Produces cliché AI filler like 'Unlock your potential!'). | "Write 3 high-converting headlines using the Problem-Solution storyline. Target: Startup founders struggling with slow websites." |
| Repository Analysis | Pasting 50 unformatted files in chat window.                                                   | Using Antigravity `@dir` context tag: "Inspect `@/components` and list unused Tailwind classes or duplicate state hooks."         |
| Debugging           | "Fix this bug." (Pasting error without code context).                                          | "Here is the terminal error stack trace: [Error]. Inspect `@app/api/contact/route.ts` and apply the smallest safe fix."           |

### YOUR TASK · 06.1 — AUDIT YOUR PROMPTING HABITS

### CHECKLIST

- ☐ 1. Do I always specify the Exact Persona & Role before requesting code?
- ☐ 2. Do I provide specific visual aesthetic references (Linear, Vercel, Apple) instead of generic terms?
- ☐ 3. Do I inspect the Deep Think reasoning steps before running generated scripts?

## 07. The 6 Levels of AI Independence & Web Architecture

Where do you stand on the AI Developer progression curve? Moving from Level 0 to Level 5 is what separates amateur prompt-copying from running a profitable AI engineering operation.

| LEVEL   | ROLE TITLE           | OPERATIONAL CAPABILITY & WORKFLOW MASTERY                                                             |
|---------|----------------------|-------------------------------------------------------------------------------------------------------|
| Level 0 | Prompt Copier        | Copies random prompts from social media without understanding frontend architecture or code output.   |
| Level 1 | UI Prompter          | Uses v0 or Bolt to generate static UI mockups in English; struggles with multi-page navigation.       |
| Level 2 | Feature Builder      | Understands Next.js App Router, Client vs Server components, and basic Supabase database integration. |
| Level 3 | Agent Debugger       | Traces terminal stack traces, reads system logs, and directs AI agents to fix complex state bugs.     |
| Level 4 | System Orchestrator  | Configures `.agents/` rules, routes tasks by model complexity, and runs Builder → Reviewer workflows. |
| Level 5 | AI Systems Architect | Directs multi-agent teams across CLI, IDE, and Browser to ship commercial SaaS and scale agencies.    |

### WEB FUNDAMENTALS: CLIENT VS SERVER ARCHITECTURE

```
CLIENT BROWSER (UI View) → [HTTPS Request] → NEXT.JS SERVER (Node Engine)
      ↑                               ↓
[State / Hydration] ← [HTML + JSON Props] ← DATABASE (Supabase / Postgres)
```

**CRITICAL ARCHITECTURE RULE:** Keep data fetching on the **Server** (React Server Components) for fast SEO performance and secure database queries. Use ``use client`` strictly for interactive UI elements (Framer Motion, modal toggles, form state).

## 08. Antigravity IDE Platform & Context Engineering (@)

Antigravity is Google DeepMind's agentic coding platform. Mastering **Context Engineering** using `@` symbols is how you give AI agents exact surgical visibility into your codebase without overwhelming the token context window.

### ANTIGRAVITY TRIPLE-PANE WORKSPACE INTERFACE:

#### 1. EXPLORER & RULES

Displays repository files, `.agents/AGENTS.md` guidelines, and active skill definitions.

#### 2. EDITOR & DIFFS

Shows live code changes with green additions and red deletions before acceptance.

#### 3. AGENT PANEL

Houses agent conversation, execution plans, terminal tasks, and subagent delegation.

### CONTEXT ENGINEERING (@) HIERARCHY LEVELS

| TAG LEVEL           | SYNTAX EXAMPLE            | USE CASE & TOKEN EFFICIENCY                                                        |
|---------------------|---------------------------|------------------------------------------------------------------------------------|
| Level 1 (Inline)    | No `@` needed             | Trivial inline edits (changing button text, altering hex color tokens).            |
| Level 2 (File)      | `@[/components/Hero.tsx]` | Targeted component refactoring while preserving existing prop signatures.          |
| Level 3 (Directory) | `@[/app/api]`             | Multi-file API route refactoring or adding new database endpoint handlers.         |
| Level 4 (Terminal)  | `@[terminal]`             | Pasting build errors or test failure logs directly for agent root-cause diagnosis. |

#### AIWEBDEV PROMPT 1.09

#### ANTIGRAVITY AGENT (CONTEXT ENGINEERING PROMPT EXAMPLE)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

Inspect `@[/app/page.tsx]` and `@[/components/Navbar.tsx]`.

Refactor the navigation links so that clicking 'Pricing' performs a smooth Framer Motion scroll to the `#pricing` element. Preserving mobile menu responsiveness.

## 08b. Lovable & Bolt: The Rapid Generation Workflow

While Antigravity is your professional engineering environment, GUI-based AI builders like **Lovable** and **Bolt.new** are incredible for 0-to-1 rapid prototyping. Use them to generate the initial foundation, then hand off to Antigravity.

### THE LOVABLE PRACTICAL WORKFLOW

1. Open **Lovable.dev** and start a new project.
2. Define your core requirements using the prompt below.
3. Preview the generated UI and inspect obvious problems.
4. Iterate using the exact iteration prompt.
5. Connect to GitHub to push the repository.

#### AIWEBDEV PROMPT 1.10

#### LOVABLE / BOLT INITIAL BUILD (PRIMARY BUILDER)

COPY THIS INTO LOVABLE/BOLT:  
Build a modern landing page for a [Niche] business.  
Architecture: Next.js + Tailwind CSS.  
Visuals: Premium dark mode, high-contrast typography, interactive Framer Motion micro-animations.  
Include a Hero section, Features Grid, and a Contact form. Ensure all components are modular and ready for backend integration later.

#### AIWEBDEV PROMPT 1.11

#### LOVABLE / BOLT ITERATION PROMPT

COPY THIS INTO LOVABLE/BOLT:  
The generation looks good, but the mobile view is broken on the features grid. Also, the hero section feels too generic.  
Please refactor the grid to be 1-column on mobile, and apply a 21st.dev-style glowing border effect to the Hero Call-to-Action button.

### V0 + 21ST.DEV COMPONENT WORKFLOW

v0 and 21st.dev are best used for **component reference work**, not entire architectures. Workflow: Find/Generate UI → Understand what you want → Bring reference into Antigravity → Adapt.

#### AIWEBDEV PROMPT 1.12

#### ANTIGRAVITY V0 COMPONENT ADAPTATION (STANDARD BUILDER)

COPY THIS INTO ANTIGRAVITY:  
I am attaching a UI component reference I generated in v0/found on 21st.dev.  
Please inspect our current project's design system (tailwind config, global css) and adapt this new component to perfectly match our existing aesthetic. Remove any generic AI-looking design patterns and ensure strict responsive behavior across mobile viewports.

## 08c. GitHub for Absolute Beginners

Git is a time machine for your code. **GitHub** is the cloud where that time machine lives. A **Repository** is your project folder. A **Commit** saves your changes. **Push** uploads them, and **Pull** downloads them. **Clone** copies a repository to your local machine.

### TRACK A: GitHub Desktop (GUI)

1. Install & Sign in to GitHub Desktop.
2. Click "Clone Repository" → select the one Lovable created.
3. Choose a local folder on your computer.
4. Open that local folder in Antigravity.
5. To save changes: Write a summary in the bottom left, click "Commit", then "Push origin".

### TRACK B: Terminal / CLI

```
'git clone [URL]' (Download repo)
'git add .' (Stage all changes)
'git commit -m "update hero"' (Save)
'git push origin main' (Upload)
'git pull' (Download latest)
```

### 🔥 ABSOLUTE RULE: NEVER COMMIT SECRETS!

Never push `.env`, `.env.local`, API keys, or private database credentials to GitHub. Always use a `.gitignore` file.

#### AIWEBDEV PROMPT 1.13

#### GITHUB PREPARATION AUDIT (QA AUDITOR)

COPY THIS INTO ANTIGRAVITY:  
Prepare this repository safely for GitHub.  
Inspect the `.gitignore` file, check for any hardcoded secret files, generated build artifacts, or environment variables. Ensure no API keys will be leaked before I commit and push.

## THE ARCHITECTURE HANDOFF (LOVABLE → ANTIGRAVITY)

When bringing a generated project into Antigravity, you must force the AI to understand the full context before it writes a single line of code.

#### AIWEBDEV PROMPT 1.14

#### REPOSITORY INSPECTION HANDOFF (SENIOR ARCHITECT)

COPY THIS INTO ANTIGRAVITY:  
I have imported this project from an AI generator (Lovable/Bolt).  
DO NOT modify anything initially.  
Run a full inspection of the repository. Identify the framework, dependencies, routing structure, components, state management, backend/DB usage, environment variables, and the design system. Document any obvious risks or technical debt. Provide a recommended list of next steps to make this production-ready.

#### AIWEBDEV PROMPT 1.15

#### MERGE CONFLICT DIAGNOSTIC (THINKING/DEBUGGING MODEL)

COPY THIS INTO ANTIGRAVITY:  
I have hit a git merge conflict. DO NOT blindly choose 'ours' or 'theirs'.  
Please inspect the conflicted files. Explain exactly what both sides (my local changes vs the incoming cloud changes) contain and what the intended behavior should be. Recommend a safe resolution strategy that preserves all intended functionality without deleting unrelated work. Verify the project builds correctly after you resolve it.

## 09. Official Model Routing Selection Matrix & Architectures (1-6)

Using the wrong AI model wastes money and degrades output quality. Use this decision matrix to route development tasks to the optimal AI model:

| TASK COMPLEXITY        | RECOMMENDED AI MODEL       | PRIMARY STRENGTHS & COST EFFICIENCY                                  |
|------------------------|----------------------------|----------------------------------------------------------------------|
| Minor UI Edits & CSS   | Gemini 3.5 Flash (Low)     | Instant response speed, minimal token cost, great for rapid styling. |
| Standard Page Building | Gemini 3.5 Flash (High)    | Primary builder model for React components and CRUD pages.           |
| Repo Architecture      | Gemini 3.1 Pro (High)      | Deep reasoning across entire multi-file project structures.          |
| Complex Debugging      | Claude Sonnet 4.6 Thinking | Senior code reviewer for logic bugs and state synchronization.       |
| Critical AppSec Audits | Claude Opus 4.6 Thinking   | High-impact security decisions, payment handlers, RLS logic.         |

### UNIVERSAL WEBSITE ARCHITECTURES (1-6)

#### 1. Business Landing Page

Lead capture, hero CTA, services, social proof, contact form. Stack: Next.js + Tailwind.

#### 2. E-Commerce Storefront

Product grid, shopping cart drawer, Stripe checkout. Stack: Next.js + Supabase + Stripe.

#### 3. Designer Portfolio

Project showcase, case study modal, MDX blog, contact form. Stack: Next.js + Framer Motion.

#### 4. Creator & Newsletter Hub

Lead magnet opt-in, bio links, latest podcast/video embeds. Stack: Next.js + Resend API.

#### 5. Blog & SEO Magazine

Category filters, search index, table of contents, author pages. Stack: Next.js + Supabase DB.

#### 6. E-Learning Course Portal

Video lesson player, progress tracker, module quiz, certificates. Stack: Next.js + Mux Video.

## 09. Universal Architectures (7-12) & Agency Niche Templates (1-5)

### UNIVERSAL WEBSITE ARCHITECTURES (7-12)

#### 7. Appointment Booking System

Interactive calendar date picker, time slot selection, SMS confirmation. Stack: Next.js + Twilio.

#### 8. Real-Time Member Community

User feeds, discussion threads, member profiles, direct messaging. Stack: Next.js + Supabase Realtime.

#### 9. Filterable Directory Engine

Multi-tag search filters, location maps, listing submissions. Stack: Next.js + Algolia / Postgres.

#### 10. SaaS Documentation Wiki

Sidebar navigation, search bar, code copy blocks, feedback widgets. Stack: Next.js + Fumadocs / MDX.

#### 11. Micro-SaaS AI Utility Tool

Prompt input textarea, API token rate limiter, result renderer. Stack: Next.js + OpenAI / Gemini API.

#### 12. B2B Client Portal Dashboard

File uploads, invoice status, project milestones, client chat. Stack: Next.js + Supabase RLS.

### TAKEIN AGENCY NICHE TEMPLATES (1-5)

#### 1. Restaurant & Café

Digital QR menu, WhatsApp table reservation CTA, Google Maps embed. Primary Goal: Instant bookings.

#### 2. Dental & Medical Clinic

Doctor profiles, patient reviews, service list, online appointment form. Primary Goal: Trust & lead capture.

#### 3. Fitness Gym & Studio

Membership plans, class schedule timetable, free trial pass signup. Primary Goal: Recurring subscriptions.

#### 4. Coaching & EdTech Academy

Course syllabus breakdown, student success videos, batch registration. Primary Goal: High ticket enrollments.

#### 5. Real Estate Brokerage

Property listings grid, filter by price/bhk, virtual tour booking. Primary Goal: High-intent buyer inquiries.

## 10. Agency Niches (6–10) & Guided Hands-on Missions

### 6. Legal & Financial Practice

Practice areas, consultation fee calculator, secure document lead form. Primary Goal: High-retainer clients.

### 7. Home Services Contractor

Emergency call CTA button, before/after photo slider, instant quote form. Primary Goal: Immediate phone leads.

### 8. Non-Profit Organization

Impact story, recurring donation integration via Stripe, volunteer signup. Primary Goal: Donor conversion.

### 9. Event Summit & Conference

Speaker lineup, agenda schedule, ticket tier pricing, sponsor logo wall. Primary Goal: Ticket sales.

## GUIDED HANDS-ON DEVELOPER MISSIONS

### LAB 01 (Guided): Hero Section Scaffold

Build a responsive Hero section with glowing pill badge using 21st.dev components. Verify desktop and mobile (375px) viewports.

### TASK 02 (Partially Guided): 3-Column Services Grid

Construct a prompt to build a 3-column Services grid with Framer Motion entry reveals and hover scale effects.

### BUILD MISSION 03: Complete Restaurant Website

Build a single-page restaurant website ("Spice & Craft") using the Golden Website Prompt Framework and deploy live.

### CHALLENGE 04: Debugging Form Persistence

The lead capture form submits but does not display success state. Direct Claude Code / Antigravity to fix the state handler!



## 11. Dual-Track Deployment Systems (GUI vs CLI)

Choose your deployment workflow based on your team's technical comfort. Both tracks ship production websites to Vercel's global CDN:

### TRACK 1: FAST PATH (0-CODE GITHUB GUI)

1. Download project ZIP from v0/Bolt export.
2. Open GitHub Desktop GUI application.
3. Select *File > Add Local Repository*.
4. Type commit summary and click **Commit to main**.
5. Click **Publish repository** to GitHub.
6. Import repo inside Vercel Dashboard for 1-click launch.

**Best For: Beginners & Non-Technical Founders**

### TRACK 2: PRO ARCHITECT (TERMINAL CLI)

1. Launch terminal in root project directory.
2. Run `'git init && git add .'` command.
3. Commit code: `'git commit -m "feat: initial commit"'`.
4. Push to remote: `'git push origin main'`.
5. Deploy directly via CLI: `'vercel --prod'`.
6. Custom domain DNS mapping via Vercel CLI.

**Best For: Full-Stack AI Engineers & Agencies**

#### AIWEBDEV PROMPT 1.10

#### TERMINAL WORKFLOW (VERCEL CLI DEPLOYMENT COMMANDS)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

```
# Step 1: Install Vercel CLI tool
npm install -g vercel
```

```
# Step 2: Authenticate and link project
vercel login
```

```
# Step 3: Deploy to production environment instantly
vercel --prod
```

#### ⚠ COMMON DEPLOYMENT MISTAKE:

Never commit `.env.local` or environment secret keys to public GitHub repositories! Always add `.env.local` to your `.gitignore` file before publishing.

## 12. Modern Design Principles & Conversion Psychology

AI can generate code, but only **Design Knowledge** tells the AI what looks world-class. Master these core design principles:

### Visual Hierarchy & Scale

- **Headline Dominance:** H1 must be at least 2.5× the size of body text.
- **Contrast:** Pure white text on deep slate background for maximum LCP readability.
- **Color Restraint:** Max 3 colors. Primary background (#0F172A), neutral (#F8F AFC), accent (#FF6B00).

### Typography & Line Height

- **Font Pair:** Inter for body and display headings. Fira Code for code/prompts.
- **Line Height:** 1.5 for body copy, tight 1.15 for display titles.
- **Tracking:** ``-0.02em`` letter spacing on display titles for modern tight look.

### THE 3 CONVERSION QUESTIONS EVERY PAGE MUST ANSWER:

1. **Why should I trust you?** → Verified client logos, real star ratings, metrics above fold.
2. **Why should I care?** → Headline speaks directly to the visitor's pain point, not company features.
3. **Why should I act NOW?** → Clear action-oriented CTA button ("Start Building Free", not "Submit").

| SECTION      | CONVERSION PRIMARY OBJECTIVE    | PROMPT SPECIFIC COMMAND                                               |
|--------------|---------------------------------|-----------------------------------------------------------------------|
| Hero         | Hook visitor in under 3 seconds | "Make H1 outcome-focused. Add trust badge below primary CTA button."  |
| Social Proof | Establish instant authority     | "Display a line of 5 recognized client brand logos with opacity 0.7." |
| Pricing      | Remove purchase friction        | "Include money-back guarantee micro-copy directly beneath CTA."       |

## 🏆 Volume 1 Capstone: Flagship SaaS Build (Part 1)

### CAPSTONE ASSIGNMENT BRIEF:

You have been hired by **"ApexAI Analytics"** (an AI startup) to design, build, and deploy their flagship web platform. This is a comprehensive multi-page handbook assignment.

#### Business Scenario & Requirements

- **Company:** ApexAI Analytics
- **Target Audience:** Enterprise CTOs & Data Leads
- **Primary Goal:** High-intent demo requests & trial signups
- **Aesthetic:** Dark Slate (#0F172A) + TakeIN Orange (#FF6B00)

#### Required Pages & Components

- Glassmorphic Navigation Bar
- Hero Section with Glowing Dashboard Mockup
- 8-Stage Conversion Layout
- Interactive Monthly/Yearly Pricing Table
- Animated FAQ Accordion + Footer

#### AIWEBDEV PROMPT 1.11

#### CAPSTONE PROMPT 1 (PHASE 1 — PLANNING & SCAFFOLD PROMPT)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

Act as Principal Architect for ApexAI Analytics.

Inspect the repository requirements. Plan a Next.js 14 App Router project structure:

1. Create `app/page.tsx` for the main high-converting landing page.
2. Create `components/ui/` for modular component primitives (Hero, Features, Pricing, Testimonials, FAQ, Footer).
3. Setup Tailwind design tokens: primary slate #0F172A, accent orange #FF6B00.
4. Output a detailed implementation plan step-by-step before writing code.

### STUDENT CAPSTONE CHECKPOINT · PHASE 1

**STATUS**

- ☐ 1. Next.js 14 project initialized with TypeScript and Tailwind CSS.
- ☐ 2. Project planning prompt executed inside Claude Code or Antigravity IDE.



## Volume 1 Capstone: Implementation & Blueprints (Part 2)

### AIWEBDEV PROMPT 1.12

### CAPSTONE PROMPT 2 (PHASE 2 – CODE BUILD & COMPONENT PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Build the full production components for ApexAI Analytics:

1. `components/ui/Hero.tsx`: Title "Turn Raw Data into Predictive Revenue", sub-headline, orange CTA button "Start Free 14-Day Trial", trust badge "Trusted by 500+ Engineering Teams". Add Framer Motion initial opacity 0, y 20 entrance animation.
2. `components/ui/Features.tsx`: 3x3 Bento grid showcasing AI Auto-Indexing, Real-Time Anomaly Detection, and Custom API Pipelines. Add hover scale 1.02x.
3. `components/ui/Pricing.tsx`: 3-tier pricing table (Starter \$49/mo, Growth \$149/mo, Enterprise \$499/mo) with interactive billing cycle toggle state.

### // ApexAI Interactive Pricing Toggle State Blueprint

```
import { useState } from 'react';
import { motion } from 'framer-motion';

export const PricingSection = () => {
  const [isYearly, setIsYearly] = useState(false);

  return (
    <section className="py-20 bg-slate-950 text-white" id="pricing">
      <div className="max-w-6xl mx-auto px-4 text-center">
        <h2 className="text-3xl font-extrabold mb-4">Flexible Pricing for Every Team</h2>
        <div className="flex items-center justify-center gap-4 mb-12">
          <span className={!isYearly ? "text-orange-500 font-bold" : "text-slate-400"}>Monthly</span>
          <button onClick={() => setIsYearly(!isYearly)} className="w-14 h-8 bg-slate-800 rounded-full p-1 border border-slate-700">
            <motion.div animate={{ x: isYearly ? 24 : 0 }} className="w-6 h-6 bg-orange-500 rounded-full" />
          </button>
          <span className={isYearly ? "text-orange-500 font-bold" : "text-slate-400"}>Yearly (Save 20%)</span>
        </div>
      </div>
    </section>
  );
};
```

### 🔥 INTENTIONAL DEBUGGING CHALLENGE:

The pricing toggle state changes in React memory, but prices fail to update visually. **Student Task:** Direct Claude Code to inspect state re-renders and fix the conditional pricing calculation!



## Volume 1 Capstone: Deployment & Final QA (Part 3)

### AIWEBDEV PROMPT 1.13

### CAPSTONE PROMPT 3 (PHASE 3 – VERIFICATION & DEPLOYMENT PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Perform a comprehensive quality audit of ApexAI Analytics:

1. Run `npm run build` and resolve any TypeScript errors or missing page key warnings.
2. Verify responsive layout rendering across 375px mobile viewports.
3. Deploy the application to Vercel production using Track 1 (GitHub Desktop GUI) or Track 2 (Vercel CLI). Output live production URL.

### STUDENT SUBMISSION & SUCCESS CRITERIA

### DELIVERABLE

- ☐ 1. Live Vercel Production URL active with custom SSL certificate.
- ☐ 2. All 8 conversion sections rendering correctly without horizontal scrollbars on mobile.
- ☐ 3. Framer Motion entrance reveals functioning smoothly on page scroll.
- ☐ 4. GitHub repository pushed with clean commit history.

PASTE YOUR LIVE VERCEL URL: \_\_\_\_\_

PASTE GITHUB REPOSITORY URL: \_\_\_\_\_

### SELF-ASSESSMENT GRADUATION SCORE

- Design Quality (Apple/Linear Vibe): \_\_\_\_ / 10
- Mobile Responsiveness (375px): \_\_\_\_ / 10
- Animation Physics (Framer Motion): \_\_\_\_ / 10
- Deployment Speed & Clean Code: \_\_\_\_ / 10

## PART 12: From Local Project to Production

### \$ The Final Frontier

Most tutorials end when the app works on your computer. But a website doesn't exist until it's live on the internet. Deployment is not just a button click; it's a critical engineering skill.

### 21. Localhost vs Production

**Localhost** is your laptop. When you run `npm run dev`, you create a tiny temporary server that only you can see. **Production** is the real world. Your code must be packaged (built), sent to a host, and connected to real databases and domains.

#### THE MENTAL MODEL

Browser (User)



Frontend (Vercel)



Backend/API  
(Render/Vercel)



Database (Supabase)

#### 🔥 CRITICAL MISCONCEPTION:

Vercel, Render, and Supabase are NOT interchangeable. Vercel is optimized for Frontend/React. Render is great for heavy Backends (like Python/FastAPI). Supabase is your Database. Don't try to host a Postgres database directly on Vercel.

## 24b. Browser DevTools for Beginners

When an AI app breaks, "It doesn't work" is not enough information for Antigravity to fix it. You must learn to open **Developer Tools (F12)**. Focus solely on two tabs:

### 1. The CONSOLE Tab

This is where JavaScript errors are logged. If a React component crashes, or a variable is undefined, it prints a red error here.

### 2. The NETWORK Tab

This records every request. **Request** (what you sent) → **Response** (what the server replied) → **Status** (200 OK, 403 Forbidden, 500 Error).

### Example Scenario:

You click "Submit". Nothing happens. You open the Network tab, click the red failed request ('POST /api/contact'), and see Status: 403. You copy this evidence for Antigravity.

AIWEBDEV PROMPT 1.15b

NETWORK FAILURE DIAGNOSTIC (DEBUGGING MODEL)

COPY THIS INTO ANTIGRAVITY:

My frontend request is failing.

In the browser Network tab, the request to `[ENDPOINT]` failed with Status Code: `[CODE]`.

The Response Payload says: `[PASTE PAYLOAD OR ERROR]`.

Please inspect the backend route and the frontend fetch call to identify the mismatch or missing permission causing this failure.

## 22. Environment Variables From Zero

Your application needs to know where the Supabase database is. But if you hardcode the database URL and passwords directly into your source code, anyone can see them on GitHub. **Environment variables** allow you to configure secrets *outside* the code.

### SERVER ONLY (SECRET)

SUPABASE\_SERVICE\_ROLE\_KEY

DATABASE\_URL

OPENAI\_API\_KEY

Never expose these to the browser.

### SAFE FOR BROWSER

NEXT\_PUBLIC\_SUPABASE\_URL

NEXT\_PUBLIC\_SUPABASE\_ANON\_KEY

The prefix NEXT\_PUBLIC\_ makes it visible.

Locally, we store these in a file called `.env.local`. We add this file to `.gitignore` so it never goes to GitHub. In production, we type these directly into Vercel's dashboard.

### AIWEBDEV PROMPT 1.14

### THE ENVIRONMENT AUDIT (DIAGNOSTIC PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

Inspect this project's environment-variable usage.  
Do not modify anything yet.

Create a table containing:

1. Variable name
2. File where it is referenced
3. What service it belongs to
4. Whether it is required
5. Browser-safe or server-only
6. Where I should obtain the value
7. Whether it needs to exist locally
8. Whether it needs to be configured in production

Do NOT print the actual values of existing secrets.

Identify missing variables and incorrectly exposed secrets. Explain everything in beginner-friendly English.



## 23. Connect Supabase Properly

Supabase provides five main tools: **PostgreSQL** (database), **Auth** (users), **Storage** (files), **RLS** (security rules), and **API clients** (communication). Let's connect it properly.

### THE CONNECTION WORKFLOW

1. Create Supabase project & wait for provisioning
2. Open project settings → API
3. Copy Project URL and anon/public key
4. Create `.env.local` in your code editor
5. Add `NEXT_PUBLIC_SUPABASE_URL=your_url`
6. Add `NEXT_PUBLIC_SUPABASE_ANON_KEY=your_key`
7. Restart your local development server! (Variables only load on start)

### AIWEBDEV PROMPT 1.15

### THE DATABASE CONNECTION (IMPLEMENTATION PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

`.env.local`

#### NEXT

Verify Implementation

I want to connect this existing application to Supabase.  
First inspect the project. Do not implement anything yet.

Determine: framework, package manager, existing Supabase dependencies, current contact form implementation, and whether a database schema already exists.

My goal is: When a visitor submits the contact form, store the enquiry securely in Supabase.  
Design the smallest appropriate implementation.

Tell me:

- what I need to configure in Supabase (SQL)
- which files you will modify
- what security considerations exist (RLS)
- how we will verify the integration

Wait for my approval before implementation.

## 24. Deploy Frontend to Vercel

Vercel is the easiest way to host a Next.js or React frontend. The golden rule is: **Vercel connects to GitHub, not to your laptop.**

### THE DEPLOYMENT PIPELINE

Local Code → Push to GitHub → Vercel Auto-Builds → Live URL

Whenever you push to the `main` branch, Vercel deploys immediately.

### "My app works locally but Vercel fails!"

This is a rite of passage. Vercel is stricter than your laptop. It checks for TypeScript errors, missing keys, and invalid imports during the **Build** phase. When a build fails, do not panic. Do this:

#### AIWEBDEV PROMPT 1.16

#### THE VERCEL BUILD FIXER (DIAGNOSTIC PROMPT)

##### USE WHEN

Following Volume I Workflow

##### CONTEXT TO ATTACH

@project

##### RUN INSIDE

Antigravity Agent

##### RECOMMENDED ROLE

Standard Builder

##### DO NOT ATTACH

.env.local

##### NEXT

Verify Implementation

My Vercel deployment failed during the build phase.  
The application works fine locally using `npm run dev`.

Here is the exact build log error from Vercel:  
[PASTE ENTIRE ERROR LOG HERE]

Please diagnose the root cause. Do not write the code fix yet.  
Explain exactly why this fails in production but not locally, and outline the steps required to resolve it.

## 25. Supabase + Vercel Environments

If you deploy to Vercel right now, your website will load but the database will fail. Why? Because `.env.local` stays on your laptop. Vercel doesn't know your Supabase URL!

### SYNCING VARIABLES TO PRODUCTION

1. Open your Vercel Dashboard → Select Project.
2. Go to **Settings** → **Environment Variables**.
3. Copy `NEXT_PUBLIC_SUPABASE_URL` from your local `.env.local` and paste it.
4. Copy `NEXT_PUBLIC_SUPABASE_ANON_KEY` and paste it.
5. Click **Save**.
6. **CRITICAL:** Vercel does NOT apply new variables to the running site automatically. You must trigger a Redeploy!

AIWEBDEV PROMPT 1.16

ENVIRONMENT VARIABLE AUDIT (QA AUDITOR)

COPY THIS INTO ANTIGRAVITY:

Please audit all environment variables required for this project.

Identify each variable, where it is used, whether it is public or a private secret, and where it must be configured (Local, Vercel, Render). Check if any are missing or pose a security risk if exposed to the frontend. DO NOT print out the actual secret values in your response.

### VERIFICATION WORKFLOW

How to prove it works:

Open Live Website → Submit Form → Open Supabase Table Editor → See Row Appear

If the row doesn't appear: Press F12 in browser, go to Console or Network tab, and look for red errors.

## Vercel Deployment Verification

AIWEBDEV PROMPT 1.17

VERCEL PRE-DEPLOYMENT QA

COPY THIS INTO ANTIGRAVITY:  
I am preparing to deploy this Next.js project to Vercel via GitHub.

AIWEBDEV PROMPT 1.18

FAILED VERCEL DEPLOYMENT DEBUG

COPY THIS INTO ANTIGRAVITY:  
My Vercel deployment failed during the build phase. The application works fine locally.

## 26. Render From Zero

If you are building a standard Next.js app, Vercel handles everything. But if your AI-built project uses a dedicated backend like **Python FastAPI** or **Node Express**, Vercel is not the right tool. You need Render.

### DEPLOYING A BACKEND TO RENDER

1. Render Dashboard → New Web Service
2. Connect GitHub Repository
3. **Build Command:** e.g., `pip install -r requirements.txt` (How it installs dependencies)
4. **Start Command:** e.g., `uvicorn main:app --host 0.0.0.0 --port 10000` (How it runs)
5. Deploy! Wait for the live URL: `https://my-api.onrender.com`

AIWEBDEV PROMPT 1.19

RENDER DEPLOYMENT PRECHECK

COPY THIS INTO ANTIGRAVITY:  
I am preparing to deploy this backend service to Render.

AIWEBDEV PROMPT 1.20

RENDER FAILURE DEBUG

COPY THIS INTO ANTIGRAVITY:  
My Render deployment failed to start, or crashed immediately after starting.

## 27. Connect Vercel Frontend → Render Backend

Locally, your frontend called `http://localhost:8000`. In production, it must call your new Render URL.

### 🔥 THE CLASSIC PRODUCTION BUG:

"Everything worked on my laptop. After deployment, the frontend buttons do nothing."

**Reason:** Your live frontend is still trying to call `localhost:8000`! You must update `NEXT_PUBLIC_API_URL=https://my-api.onrender.com` in Vercel.

AIWEBDEV PROMPT 1.21

FRONTEND BACKEND CONNECTION DIAGNOSTIC

COPY THIS INTO ANTIGRAVITY:

## 28. CORS Without the Headache

When your Vercel frontend (Website A) tries to talk to your Render backend (Website B), the browser blocks it by default. This security feature is called **CORS** (Cross-Origin Resource Sharing). Your backend must explicitly give permission.

### AIWEBDEV PROMPT 1.17

### CORS CONFIGURATOR (IMPLEMENTATION PROMPT)

#### USE WHEN

Following Volume I Workflow

#### CONTEXT TO ATTACH

@project

#### RUN INSIDE

Antigravity Agent

#### RECOMMENDED ROLE

Standard Builder

#### DO NOT ATTACH

.env.local

#### NEXT

Verify Implementation

My frontend is hosted at `https://my-app.vercel.app` and my backend is hosted on Render. I am getting a CORS error in the browser console when calling the API.

Inspect my backend code and configure CORS properly so it accepts requests from the production frontend URL, and `http://localhost:3000` for local development. Explain how you are configuring it securely.

## 29. Full Environment Map

Reference this table when debugging missing configuration:

| Variable     | Local          | Vercel      | Render      | Public?   |
|--------------|----------------|-------------|-------------|-----------|
| Supabase URL | ✓              | ✓           | Maybe       | Depends   |
| Anon Key     | ✓              | ✓           | Maybe       | Yes (RLS) |
| Service Key  | Only if server | Server Only | Server Only | NO        |
| API_URL      | ✓              | ✓           | —           | Yes       |

### 30. The “Environment Variables Are Broken” Mission

Theory is useless until you fix a broken system. This is a practical challenge. Your current local repository has a broken configuration. The website loads, but the form fails. Data never reaches Supabase.

#### YOUR MISSION OBJECTIVE

#### ACTIVE TRACING

Use Claude Code or Antigravity to trace the data flow from the browser all the way to Supabase. You must find the missing link.

- 1. Inspect the form submit function (Browser).
- 2. Check which environment variable it uses for the API.
- 3. Check the API route/backend handler.
- 4. Verify the database credentials being injected into the backend.
- 5. Fix the ``.env.local`` to successfully push data to Supabase.

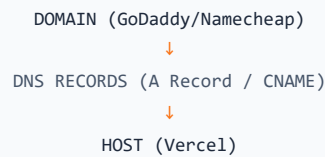
#### PRO-TIP: DON'T GUESS. VERIFY.

When something breaks, AI will often hallucinate complex code rewrites. Tell the AI: "Stop. Do not write code. Tell me exactly what the value of `NEXT_PUBLIC_API_URL` is resolving to right now."



## 31. Domains & DNS From Zero

my-app.vercel.app is nice, but you want takeinstudio.com. DNS (Domain Name System) is simply the internet's phonebook. It tells traffic where your code lives.



## 32. GoDaddy → Vercel Workflow

Do not try to memorize Vercel's IP address. Always follow Vercel's UI.

1. **In Vercel:** Settings → Domains → Add 'yourdomain.com'.
2. Vercel will give you an A Record (IP address) and a CNAME.
3. **In GoDaddy:** DNS Settings → Add the exact records Vercel showed you.
4. **Wait:** DNS propagation can take 5 minutes to 24 hours.
5. **In Vercel:** Wait for the SSL certificate to generate automatically.

AIWEBDEV PROMPT 1.22

DNS CONFIGURATION ASSISTANT

COPY THIS INTO ANTIGRAVITY:  
I am connecting a custom domain to Vercel.  
Vercel is asking me to add these exact records:  
[PASTE VERCEL DNS REQUIREMENTS HERE]  
My domain registrar is [e.g. GoDaddy/Namecheap]. Can you tell me exactly what to click and type in my registrar's DNS settings panel to configure this? Do not hardcode values; interpret the ones I pasted.

AIWEBDEV PROMPT 1.23

DOMAIN NOT VERIFYING DEBUG

COPY THIS INTO ANTIGRAVITY:  
It has been 2 hours since I updated my DNS records, but Vercel still says "Invalid Configuration".  
I entered the following records at my registrar:  
[PASTE RECORDS YOU ENTERED]  
Can you verify if I made a mistake with the CNAME apex/www rules, or if it is just normal DNS propagation delay?

### 33. Backend Subdomain Architecture

For advanced projects, you split traffic using subdomains.  
`www.takeinstudio.com` points to Vercel (Frontend).  
`api.takeinstudio.com` points to Render (Backend).

### 34. Production Debugging With AI

The most important prompt format for deployment failures:

**AIWEBDEV PROMPT 1.18****THE PRODUCTION TRIAGE (DIAGNOSTIC PROMPT)**

|                             |                       |
|-----------------------------|-----------------------|
| USE WHEN                    | RECOMMENDED ROLE      |
| Following Volume I Workflow | Standard Builder      |
| CONTEXT TO ATTACH           | DO NOT ATTACH         |
| @project                    | .env.local            |
| RUN INSIDE                  | NEXT                  |
| Antigravity Agent           | Verify Implementation |

My application is deployed to production, but I am facing an issue.  
[Frontend is on Vercel, Backend is on Render, Database is Supabase].

Symptom: [e.g., 500 server error when logging in / form doesn't submit / CORS error in console].

Here are the deployment logs / console errors:  
[PASTE LOGS]

1. Where exactly in the chain (Browser → Frontend → Backend → DB) is the break occurring?
2. What specific context or files do you need me to show you to diagnose this further?
3. What environment variables should I double-check in Vercel/Render?

Do not guess the solution. Walk me through the diagnostic process.



## 34b. Volume I Final Mission: The End-to-End Build

You must now execute everything you've learned. Do not use one giant prompt. Follow this staged workflow.

### STAGE 1: GENERATION

**LOVABLE / BOLT**

Research a local business. Use the Golden Website Prompt to generate their landing page in an AI GUI builder. Iterate once to fix obvious UI flaws.

### STAGE 2: HANDOFF & GITHUB

**GITHUB DESKTOP**

Connect the repository to GitHub. Clone it locally using GitHub Desktop. Open it in Antigravity and run the Repository Inspection Handoff prompt.

### STAGE 3: LOCAL ITERATION

**ANTIGRAVITY**

Add a complex 21st.dev interactive component. Run ``npm run dev`` and test it in your browser. Inspect the Console tab for any warnings.

### STAGE 4: DEPLOYMENT

**VERCEL + DNS**

Run the Vercel Pre-Deployment prompt. Push to GitHub to trigger Vercel deployment. Add a custom domain (or subdomain) and follow the DNS Configuration steps.

### STAGE 5: FINAL QA

**MOBILE BROWSER**

Open your live custom domain on your actual phone. Verify touch targets, margins, and the Vercel logs.



## 35. Volume 1 Assessment Checkpoint & Graduation

### CONCEPTUAL MASTERY CHECKPOINT — ANSWER WITHOUT LOOKING BACK

1. Explain the fundamental difference between Claude Code CLI and standard ChatGPT chat UI:

---

2. Why should heavy animations never exceed 0.6 seconds in duration on commercial web pages?

---

3. State the 5 core pillars of the Golden Website Prompt Framework:

---

4. What is the operational difference between Gemini 3.5 Flash and Claude Sonnet 4.6 Thinking?

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### OFFICIAL VOLUME 1 GRADUATION CERTIFICATE

#### AI-Native Developer Foundations Mastered

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